

Exercise 4.1 Class 9



Exercise 4.1

Factorize the following

(i), (ii), (iii), (iv), (v), (vi)



(i)

$$4x + 16y + 24z$$

Solution:

Since, $ab + ac = a(b + c)$

Here, common factor = 4

Answer: $4(x + 4y + 6z)$

(ii)

$$x^2 + 3x^2y + 4x^2y^2z$$

Solution:

Since, $ab + ac = a(b + c)$

Here, common factor = x^2

Answer: $x^2(1 + 3y + 4y^2z)$

(iii)

$$3pqr + 6pqt + 3pqs$$

Solution:

Since, $ab + ac = a(b + c)$

Here, common factor = $3pq$

Answer:

$$3pq(r + 2t + s)$$

(iv)

$$9qr(s^2 + t^2) + 18q^2r^2(s^2 + t^2)$$

Solution:

Since, $ab + ac = a(b + c)$

Here, common factor = $9qr(s^2 + t^2)$

Answer:

$$9qr(s^2 + t^2)(1 + 2qr)$$

(v)

$$\frac{z^2x}{16} - \frac{x^2z^2}{8} + \frac{x^2z^3}{12}$$

Solution:

Since, $ab + ac = a(b + c)$

Here, common factor = $\frac{xz^2}{48}$

Answer: $\frac{xz^2}{48}(3 - 6x + 4xz)$

(vi)

$$a(x - y) - a^2b(x - y) + a^2b^2(x - y)$$

Solution:

Since, $ab + ac = a(b + c)$

Here, common factor = $a(x - y)$

Answer: $a(x - y)(1 - ab + ab^2)$

Exercise 4.1

Factorize the following

(i), (ii), (iii), (iv)



(i)

$$7x + xz + 7z + z^2$$

Solution:

Since, $ax + ay + bx + by = (x + y)(a + b)$

$$(7x + xz) + (7z + z^2)$$

$$x(7 + z) + z(7 + z)$$

Answer:

$$(x + z)(z + 7)$$

(ii)

$$9a^2b + 18ab^2 - 6ac - 12bc$$

Solution:

Since, $ax + ay + bx + by = (x + y)(a + b)$

$$(9a^2b + 18ab^2) - (6ac + 12bc)$$

$$9ab(a + 2b) - 6c(a + 2b)$$

Answer:

$$3(a + 2b)(3ab - 2c)$$

(iii)

$$6t - 12p + 4tq - 8pq$$

Solution:

Since, $ax + ay + bx + by = (x + y)(a + b)$

$$(6t - 12p) + (4tq - 8pq)$$

$$6(t - 2p) + 4q(t - 2p)$$

Answer:

$$2(t - 2p)(2q + 3)$$

(iv)

$$r^2 + 9rs - 7rs - 63s^2$$

Solution:

Since, $ax + ay + bx + by = (x + y)(a + b)$

$$(r^2 + 9rs) - (7rs + 63s^2)$$

$$r(r + 9s) - 7s(r + 9s)$$

Answer:

$$(r + 9s)(r - 7s)$$

Exercise 4.1

Find the factors of

(i), (ii), (iii), (iv), (v), (vi)



(i)

$$4a^2 + 12ab + 9b^2$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

Here, $a = 2a$, $b = 3b$

Answer:

$$(2a + 3b)^2$$

(ii)

$$36x^4 + 12x^2 + 1$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

Here, $a = 6x^2$, $b = 1$

Answer:

$$(6x^2 + 1)^2$$

(iii)

$$x^2 + 1 + \frac{1}{4x^2}$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

Here, $a = x$, $b = \frac{1}{2x}$

Answer: $\left(x + \frac{1}{2x}\right)^2$

(iv)

$$81y^2 + 144yz + 64z^2$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

Here, $a = 9y$, $b = 8z$

Answer: $(9y + 8z)^2$

(v)

$$625 + 50a^2b + a^4b^2$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

Here, $a = 25$, $b = a^2b$

Answer: $(25 + a^2b)^2$

(vi)

$$a^2 + 0.4a + 0.04$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

Here, $a = a$, $b = 0.2$

Answer: $(a + 0.2)^2$

Exercise 4.1

Factorize

(i), (ii), (iii), (iv), (v), (vi)



(i)

$$b^4 - 4b^2c^2 + 4c^4$$

Solution:

Since, $a^2 - 2ab + b^2 = (a - b)^2$

Here, $a = b^2$, $b = 2c^2$

Answer:

$$(b^2 - 2c^2)^2$$

(ii)

$$\frac{9}{4}x^4 - 2 + \frac{4}{9x^4}$$

Solution:

Since, $a^2 - 2ab + b^2 = (a - b)^2$

Here, $a = \frac{3x^2}{2}$, $b = \frac{2}{3x^2}$

Answer:

$$\left(\frac{3x^2}{2} - \frac{2}{3x^2}\right)^2$$

(iii)

$$2a^3b^3 - 16a^2b^4 + 32ab^5$$

Solution:

Since, $ab + ac = a(b + c)$

Here, common factor = $2ab^3 \Rightarrow 2ab^3(a^2 - 8ab + 16b^2)$

Since, $a^2 - 2ab + b^2 = (a - b)^2 \Rightarrow a = a, b = 4b$

Answer:

$$2ab^3(a - 4b)^2$$

(iv)

$$9(p + q)^2 - 6(p + q)r^2 + r^4$$

Solution:

Since, $a^2 - 2ab + b^2 = (a - b)^2$

Here, $a = 3(p + q), b = r^2$

Answer:

$$(3p + 3q - r^2)^2$$

(v)

$$x^2y^2 - 0.1xy + 0.0025$$

Solution:

Since, $a^2 - 2ab + b^2 = (a - b)^2$

Here, $a = xy$, $b = 0.05$

Answer: $(xy - 0.05)^2$

(vi)

$$(a - b)^2 - 18(a - b) + 81$$

Solution:

Since, $a^2 - 2ab + b^2 = (a - b)^2$

Here, $a = (a - b)$, $b = 9$

Answer: $(a - b - 9)^2$

Exercise 4.1

Factorize

(i), (ii), (iii), (iv), (v), (vi)



(i)

$$4a^2 - 9b^2$$

Solution:

$$\text{Since, } a^2 - b^2 = (a - b)(a + b)$$

$$\text{Here, } a = 2a, \quad b = 3b$$

Answer: $(2a - 3b)(2a + 3b)$

(ii)

$$16x^2 - 25y^2$$

Solution:

$$\text{Since, } a^2 - b^2 = (a - b)(a + b)$$

$$\text{Here, } a = 4x, \quad b = 5y$$

Answer: $(4x - 5y)(4x + 5y)$

(iii)

$$100x^2z^2 - y^4$$

Solution:

$$\text{Since, } a^2 - b^2 = (a - b)(a + b)$$

$$\text{Here, } a = 10xz, \quad b = y^2$$

Answer: $(10xz - y^2)(10xz + y^2)$

(iv)

$$\frac{1}{100}x^4 - 100y^4$$

Solution:

$$\text{Since, } a^2 - b^2 = (a - b)(a + b)$$

$$\text{Here, } a = \frac{x^2}{10}, \quad b = 10y^2$$

Answer: $\left(\frac{x^2}{10} - 10y^2\right)\left(\frac{x^2}{10} + 10y^2\right)$

(v)

$$\frac{64}{81}f^2 - \frac{81}{64}g^4$$

Solution:

$$\text{Since, } a^2 - b^2 = (a - b)(a + b)$$

$$\text{Here, } a = \frac{8f}{9}, \quad b = \frac{9g^2}{8}$$

$$\text{Answer: } \left(\frac{8f}{9} - \frac{9g^2}{8}\right)\left(\frac{8f}{9} + \frac{9g^2}{8}\right)$$

(vi)

$$\frac{x^4}{121} - 121y^2$$

Solution:

$$\text{Since, } a^2 - b^2 = (a - b)(a + b)$$

$$\text{Here, } a = \frac{x^2}{11}, \quad b = 11y$$

$$\text{Answer: } \left(\frac{x^2}{11} - 11y\right)\left(\frac{x^2}{11} + 11y\right)$$

Exercise 4.1

Find the factors of

(i), (ii), (iii), (iv), (v), (vi)



(i)

$$(2x + z)^2 - (2x - z)^2$$

Solution:

Since, $a^2 - b^2 = (a - b)(a + b)$

Here, $a = (2x + z)$, $b = (2x - z)$

$$a - b = 2z, \quad a + b = 4x$$

Answer:

$$8xz$$

(ii)

$$(4a - 9b)^2 - (2a + 5b)^2$$

Solution:

Since, $a^2 - b^2 = (a - b)(a + b)$

Here, $a = (4a - 9b)$, $b = (2a + 5b)$

$$a - b = 2(a - 7b), \quad a + b = 2(3a - 2b)$$

Answer:

$$4(a - 7b)(3a - 2b)$$

(iii)

$$169x^4 - (3t + 4)^2$$

Solution:

$$\text{Since, } a^2 - b^2 = (a - b)(a + b)$$

$$\text{Here, } a = 13x^2, \quad b = (3t + 4)$$

Answer: $(13x^2 - 3t - 4)(13x^2 + 3t + 4)$

(iv)

$$(9x^2 - 4y^2)^2 - (4x^2 - y^2)^2$$

Solution:

$$\text{Since, } a^2 - b^2 = (a - b)(a + b)$$

$$\text{Here, } a = (9x^2 - 4y^2), \quad b = (4x^2 - y^2)$$

$$a - b = 5x^2 - 3y^2, \quad a + b = 13x^2 - 5y^2$$

Answer: $(5x^2 - 3y^2)(13x^2 - 5y^2)$

(v)

$$\left(a^2 + 2 + \frac{1}{a^2}\right) - \left(b^2 - 2 + \frac{1}{b^2}\right)$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$ and $a^2 - 2ab + b^2 = (a - b)^2$

$$\left(a + \frac{1}{a}\right)^2 - \left(b - \frac{1}{b}\right)^2$$

Since, $a^2 - b^2 = (a - b)(a + b)$

Answer: $\left(a + \frac{1}{a} - b + \frac{1}{b}\right)\left(a + \frac{1}{a} + b - \frac{1}{b}\right)$

(vi)

$$9x^2 + \frac{1}{9x^2} - 4y^2 - \frac{1}{4y^2} + 4$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$ and $a^2 - 2ab + b^2 = (a - b)^2$

$$\left(3x + \frac{1}{3x}\right)^2 - \left(2y - \frac{1}{2y}\right)^2$$

Since, $a^2 - b^2 = (a - b)(a + b)$

Answer: $\left(3x + \frac{1}{3x} - 2y + \frac{1}{2y}\right)\left(3x + \frac{1}{3x} + 2y - \frac{1}{2y}\right)$

Exercise 4.1

Find the factors of

(i), (ii)



(i)

$$(x^2 + 2xy + y^2) - 9z^4$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

$$(x + y)^2 - (3z^2)^2$$

Since, $a^2 - b^2 = (a - b)(a + b)$

Answer:

$$(x + y - 3z^2)(x + y + 3z^2)$$

(ii)

$$(4a^2 + 8ab^2 + 4b^4) - 9c^2$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

$$(2a + 2b^2)^2 - (3c)^2$$

Since, $a^2 - b^2 = (a - b)(a + b)$

Answer:

$$(2a + 2b^2 - 3c)(2a + 2b^2 + 3c)$$